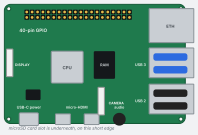
RAQIB · AI INSPECTION ROBOT · RELIABLE BUDGET BUILD

Parts & Wiring Atlas

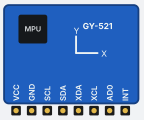
Every sensor and board in colour with its name and real size, the Raspberry Pi pin by pin, the robot's structure layer by layer, and how each part is connected.



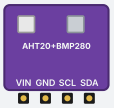
Raspberry Pi 4



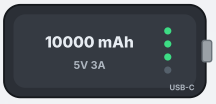
HC-SR04



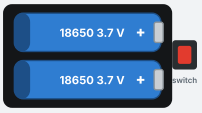
MPU6050



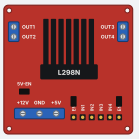
AHT20 + BMP280



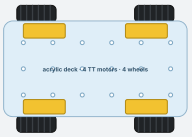
Power bank



Motor battery



L298N



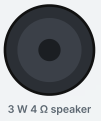
4WD chassis



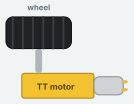
USB webcam



PAM8403



Speaker



TT motor

Ahmed Mahram Saeed Saeed Hadi
Final-year IoT project · UNISIRAJ

Atlas 1.1 · October 2026
Matches the Raqib software 2.5 and the build guide

Inside this atlas

1. **Every part** - 19 parts in six colour groups, each with a big picture, its name, how to recognise it and where its pins go
2. **The boards** - the Raspberry Pi 4 and its ports, all 40 pins, and close-ups of the L298N, HC-SR04, power bank, MPU6050, AHT20 and the hub board
3. **The robot's structure** - exploded view, side, top, bottom and front views
4. **How everything connects** - the system map, one wiring diagram per group, the complete harness and a wire-by-wire table

About the pictures: they are drawings of the common versions of each board, in their real colours and layout. Your board may be a slightly different colour or have its pins in another order: **always read the labels printed on the board**.

Wire colours used in every diagram

5 V · 3.3 V · GND · SDA · SCL · battery + · motor · PWM speed · signal · audio

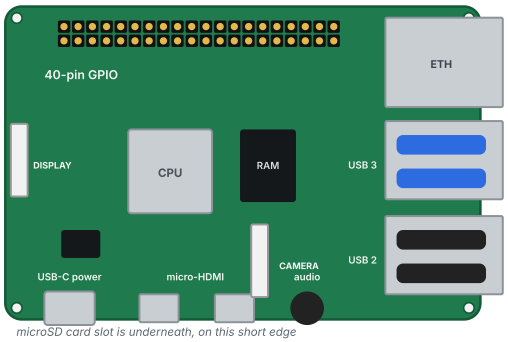
Use the same colours for your real wires where you can: mistakes become easy to spot.

Colour groups: The brain · Power · Moving · Sensing · Camera and sound · Wiring and building

The brain

The computer and its memory card.

1 Raspberry Pi 4 Model B (2 GB)



microSD card slot is underneath, on this short edge

WHAT IT DOES

The robot's computer. Runs the robot program, its own server for the phone app, and talks to the AI.

HOW TO RECOGNISE IT

Green board the size of a credit card. Silver USB and Ethernet ports on one short edge, 40 gold pins along one long edge.

REAL SIZE

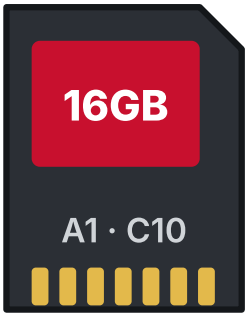
85 × 56 mm

CONNECTIONS

40-pin header	all sensors and the L298N (see the pin map)
USB-C	5 V from the power bank (its own cable)
USB 3 (blue)	USB webcam
3.5 mm jack	PAM8403 amplifier

💡 The microSD card slot is underneath, on the short edge without ports.

2 microSD card 16 GB



WHAT IT DOES

Holds Raspberry Pi OS and all the robot software.

HOW TO RECOGNISE IT

Fingernail-size card; buy 'A1' or 'Class 10'.

REAL SIZE

15 × 11 mm

CONNECTIONS

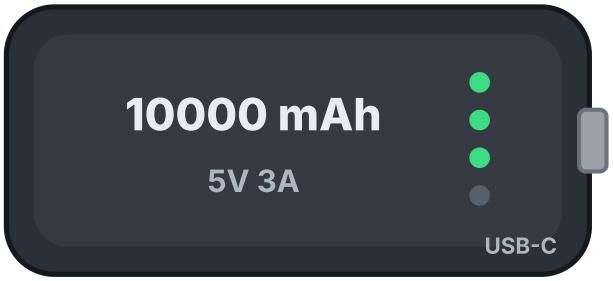
card	slot under the Pi

💡 Write it with Raspberry Pi Imager on your PC.

Power

Two separate batteries: a USB power bank for the Pi, a 2 × 18650 pack for the motors. One shared ground.

3 USB power bank 10,000 mAh (5 V 3 A)



WHAT IT DOES

Powers the Raspberry Pi, and nothing else: motor starts can never restart it.

HOW TO RECOGNISE IT

Flat black or white box with 4 small LEDs, a button, and USB-A / USB-C ports.

REAL SIZE

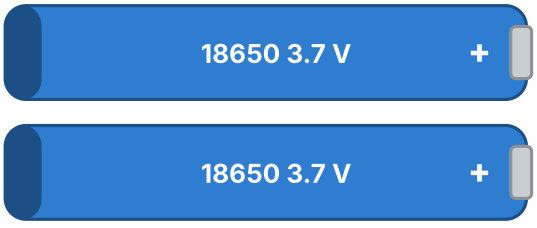
about 140 × 68 × 16 mm

CONNECTIONS

● **USB-C (or USB-A) out**	its own short cable → Pi USB-C power
● **LEDs**	how full it is (no wires)

💡 It must say 5 V 3 A (15 W) or more. Charge it like a phone.

4 18650 Li-ion cells (×2)



WHAT IT DOES

Store the energy for the motors.

HOW TO RECOGNISE IT

Cylinder like a fat AA battery, usually blue, green or purple wrapping, flat or button top.

REAL SIZE

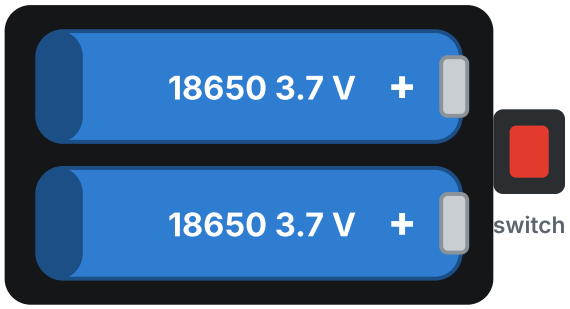
18 mm × 65 mm each, 3.7 V

CONNECTIONS

● **+ (button end)**	+ mark in the holder
● **– (flat end)**	spring in the holder

💡 Real cells are 2500-3500 mAh. '9900 mAh' cells are fake and unsafe.

5 2 × 18650 holder with switch



WHAT IT DOES
Joins the two cells into a 7.4 V motor pack; its switch turns the motors on and off.

HOW TO RECOGNISE IT
Black plastic box with two slots, a small slide switch and a red + black lead.

REAL SIZE
about 77 × 41 mm

CONNECTIONS

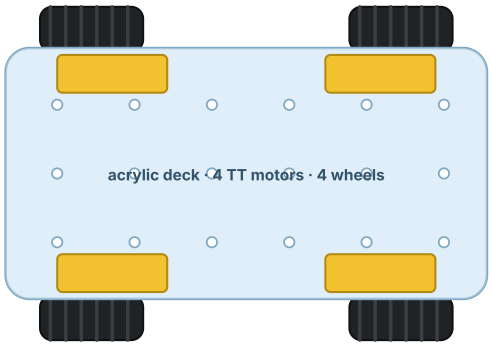
● **red lead +**	L298N +12V
● **black lead –**	L298N GND (which also goes to Pi pin 9)

💡 Full = 8.4 V, empty ≈ 6.6 V. Never connect its + to the Pi.

Moving

Body, wheels and the motor driver.

6 4WD chassis kit



WHAT IT DOES

The robot's body: two acrylic decks, four motors with wheels, brass standoffs.

HOW TO RECOGNISE IT

Two clear or coloured acrylic plates with many holes, 4 yellow TT motors, 4 rubber wheels, brass posts and screws.

REAL SIZE

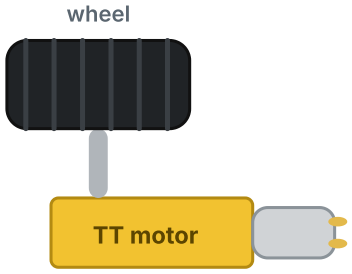
about 25 × 15 cm

CONNECTIONS

● **4 motors**	under the bottom deck, 2 per side
● **standoffs**	bottom deck → top deck

💡 Four driven wheels steer like a tank: left pair and right pair at different speeds.

7 TT gear motors (×4)



WHAT IT DOES

Each turns one wheel. The robot knows how far it went from motor power × time (calibrated once).

HOW TO RECOGNISE IT

Yellow plastic gearbox with a small metal motor; shaft out of both sides.

REAL SIZE

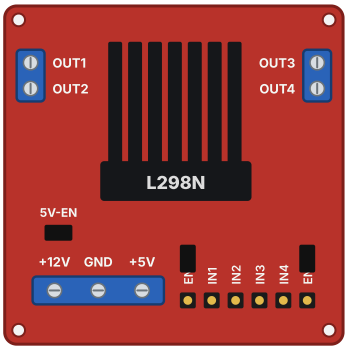
70 × 22 × 18 mm

CONNECTIONS

● **left pair red / black**	L298N OUT1 / OUT2 (front + rear joined)
● **right pair red / black**	L298N OUT3 / OUT4

💡 Join the two motors of a side red-to-red, black-to-black; a side that runs backwards: swap its two wires.

8 L298N motor driver



WHAT IT DOES

Switches motor-battery power to the motors forwards or backwards, at the speed the Pi asks.

HOW TO RECOGNISE IT

Red board with a big black finned heatsink, blue screw terminals on three sides, 6 pins on the front.

REAL SIZE

43 × 43 mm

CONNECTIONS

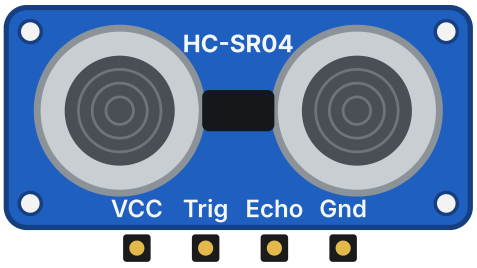
+12V / GND	motor battery + / –
GND	also to Pi pin 9: COMMON GROUND (essential)
OUT1/OUT2	left motor pair
OUT3/OUT4	right motor
ENA / ENB	Pi pins 33 / 35 (speed)
IN1-IN4	Pi pins 11, 13, 15, 16 (direction)

💡 Pull off the ENA and ENB jumpers. Keep the 5V-EN jumper.

Sensing

Distance, turning and climate sensors.

9 HC-SR04 ultrasonic sensor (×3)



WHAT IT DOES

Measures distance with sound (2 cm-4 m): stops the robot before obstacles.

HOW TO RECOGNISE IT

Blue board with two silver 'eyes' (speaker and microphone), 4 pins at the bottom.

REAL SIZE

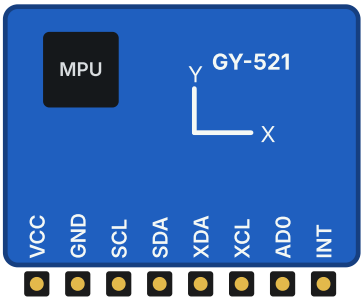
45 × 20 mm

CONNECTIONS

VCC	hub 5V
Trig	Pi pin 32 / 18 / 38 (left / centre / right)
Echo	1k/2k divider → Pi pin 36 / 22 / 40
Gnd	hub GND

💡 Echo is 5 V: never connect it to the Pi without the divider. Buy one spare.

10 MPU6050 motion sensor (GY-521)



WHAT IT DOES

Gyroscope + accelerometer: knows how far the robot has turned and keeps it straight.

HOW TO RECOGNISE IT

Tiny blue board with a square black chip and 8 pins in a row.

REAL SIZE

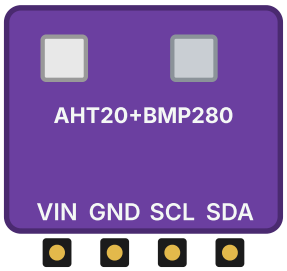
21 × 16 mm

CONNECTIONS

VCC	hub 3V3
GND	hub GND
SCL	hub SCL (Pi pin 5)
SDA	hub SDA (Pi pin 3)

💡 Mount it flat near the middle; I²C address 0x68.

11 AHT20 + BMP280 climate sensor



WHAT IT DOES
Temperature, humidity (AHT20) and air pressure (BMP280) on one board.

HOW TO RECOGNISE IT
Small purple or blue board with a white square chip (AHT20) and a tiny silver one (BMP280).

REAL SIZE
about 15 × 12 mm

CONNECTIONS

VIN	hub 3V3
GND	hub GND
SCL	hub SCL
SDA	hub SDA

💡 Two I²C addresses: 0x38 and 0x77 (or 0x76). Keep it away from warm parts.

Camera and sound

Seeing, hearing and speaking.

12 USB webcam with microphone



WHAT IT DOES
The robot's eyes and ears: photos, live video and sound.

HOW TO RECOGNISE IT
Small black camera on a clip, one USB cable, tiny mic holes on the front.

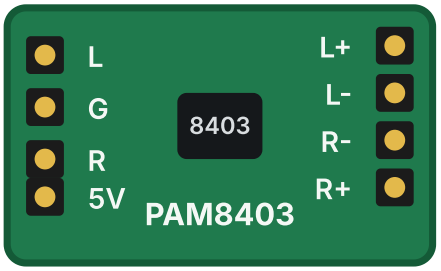
REAL SIZE
about 70-90 mm wide

CONNECTIONS

USB plug	Pi USB 3 port (blue)

💡 Choose 'UVC / plug and play' with a built-in mic. It is fixed: the robot turns to look.

13 PAM8403 amplifier



WHAT IT DOES
Makes the Pi's audio loud enough for the speaker.

HOW TO RECOGNISE IT
Tiny green board (thumbnail size) with a small chip and pins or pads.

REAL SIZE
about 21 × 18 mm

CONNECTIONS

5V / GND	hub 5V / GND
L / G	3.5 mm cable: tip / shield
L+ / L−	speaker

💡 Use only the left channel.

14 Speaker 3 W 4 Ω



3 W 4 Ω speaker

WHAT IT DOES
The robot's voice.

HOW TO RECOGNISE IT
Round, about 40 mm, black cone, two solder tabs.

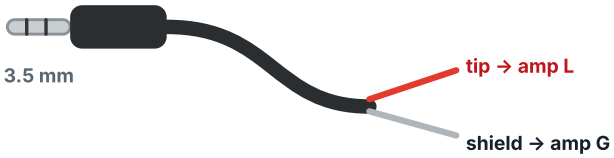
REAL SIZE
40 mm

CONNECTIONS

+ tab	PAM8403 L+
– tab	PAM8403 L–

💡 Face it outwards at the back of the top deck.

15 3.5 mm audio cable (one end cut)



WHAT IT DOES
Takes the sound from the Pi's audio jack to the amplifier.

HOW TO RECOGNISE IT
Ordinary headphone-plug cable.

REAL SIZE
20-30 cm

CONNECTIONS

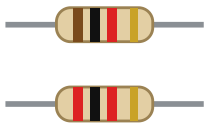
plug	Pi 3.5 mm audio jack
tip wire	PAM8403 L
shield	PAM8403 G

💡 If you are unsure which wire is the tip, test with the multimeter (continuity).

Wiring and building

What holds the robot and its wires together.

16 Resistors (1 kΩ ×3, 2 kΩ ×3)



1 kΩ brown-black-red (x 3)

2 kΩ red-black-red (x 3)

WHAT IT DOES

Three voltage dividers: they protect the Pi from the sonars' 5 V Echo.

HOW TO RECOGNISE IT

Small beige or blue tubes with coloured bands.

REAL SIZE

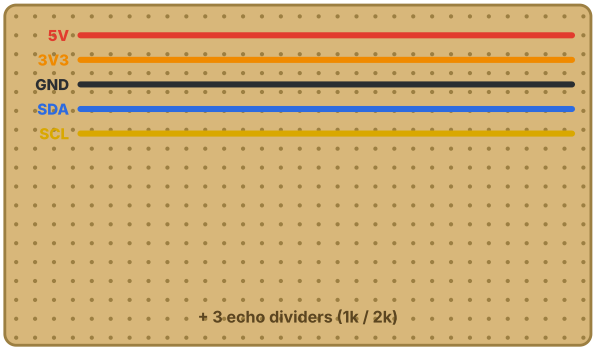
about 6 mm long

CONNECTIONS

🟢 **1 kΩ (top)**	sonar Echo → middle point
⚫ **2 kΩ (bottom)**	middle point → GND

💡 No 2 kΩ? Two 1 kΩ in series. Read the colour bands (picture).

17 Hub board (you make it)



WHAT IT DOES

One place where 5 V, 3.3 V, GND, SDA and SCL meet, plus the 3 sonar dividers.

HOW TO RECOGNISE IT

Brown perfboard with five soldered rails and header strips.

REAL SIZE

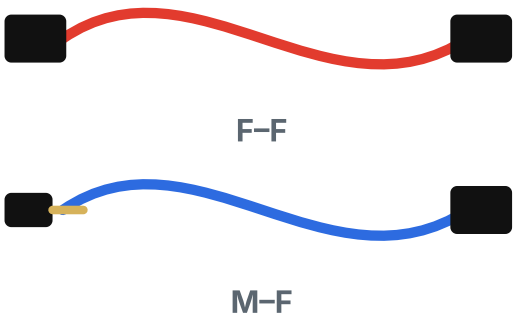
about 7 × 9 cm

CONNECTIONS

🔴 **5V / 3V3 / GND**	Pi pins 2 / 1 / 6
🔵 **SDA / SCL**	Pi pins 3 / 5

💡 Label each rail with a marker.

18 Jumper wires



WHAT IT DOES
Plug-in connections: no soldering for most parts.

HOW TO RECOGNISE IT
Rainbow ribbon of wires; F = socket end, M = pin end.

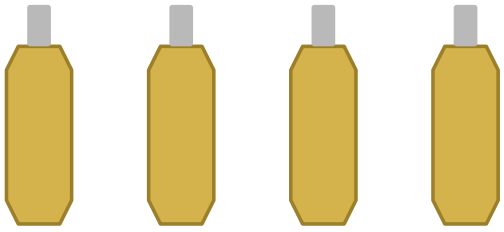
REAL SIZE
10-20 cm

CONNECTIONS

F-F	pin to pin (Pi to sensor)
M-F	pin to socket

💡 Use red for 5 V, orange for 3.3 V, black for GND.

19 Standoffs and screws



M3 standoffs 35-40 mm + screws

WHAT IT DOES
Hold the top deck above the bottom deck, and the boards on the decks.

HOW TO RECOGNISE IT
Brass or nylon hexagon posts with threads.

REAL SIZE
M3, 35-40 mm long

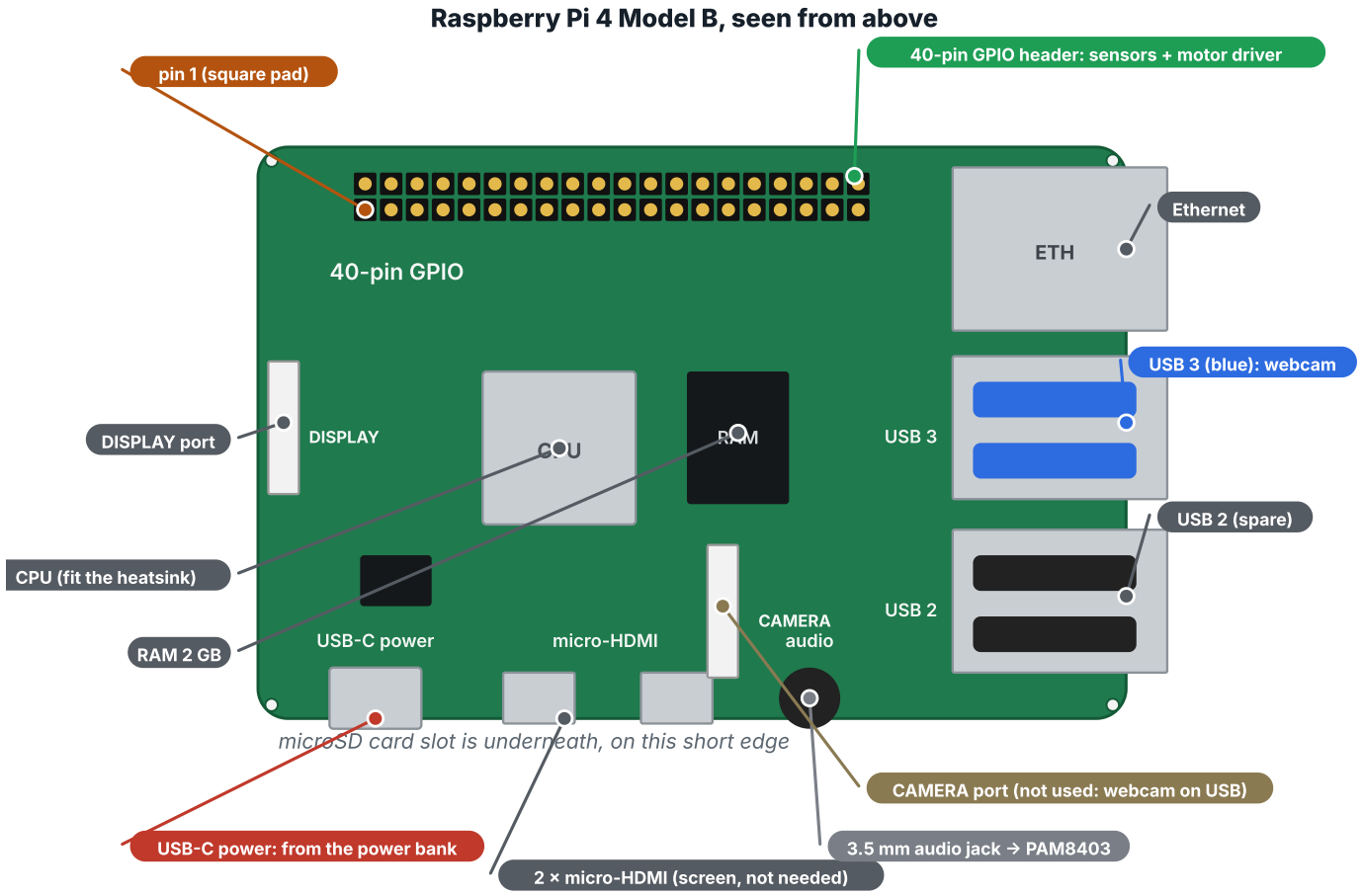
CONNECTIONS

M3 posts	between the decks
M2.5 screws	Raspberry Pi

💡 Nylon washers under boards stop short circuits.

B The boards

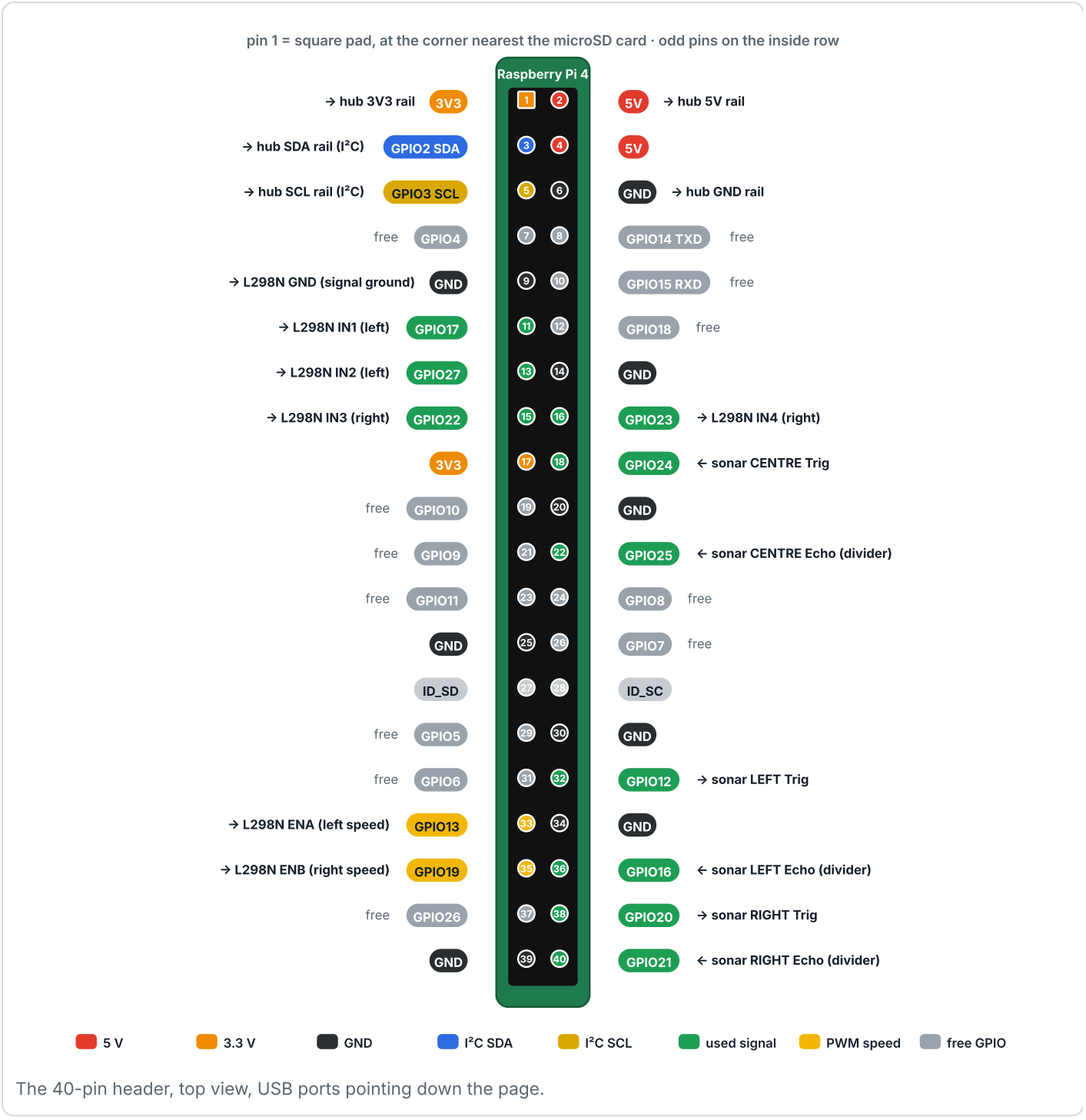
The Raspberry Pi 4 is the centre of everything: every other part connects to its pins or ports.



The Raspberry Pi 4 and what each port is used for in this robot.

All 40 pins of the Raspberry Pi header

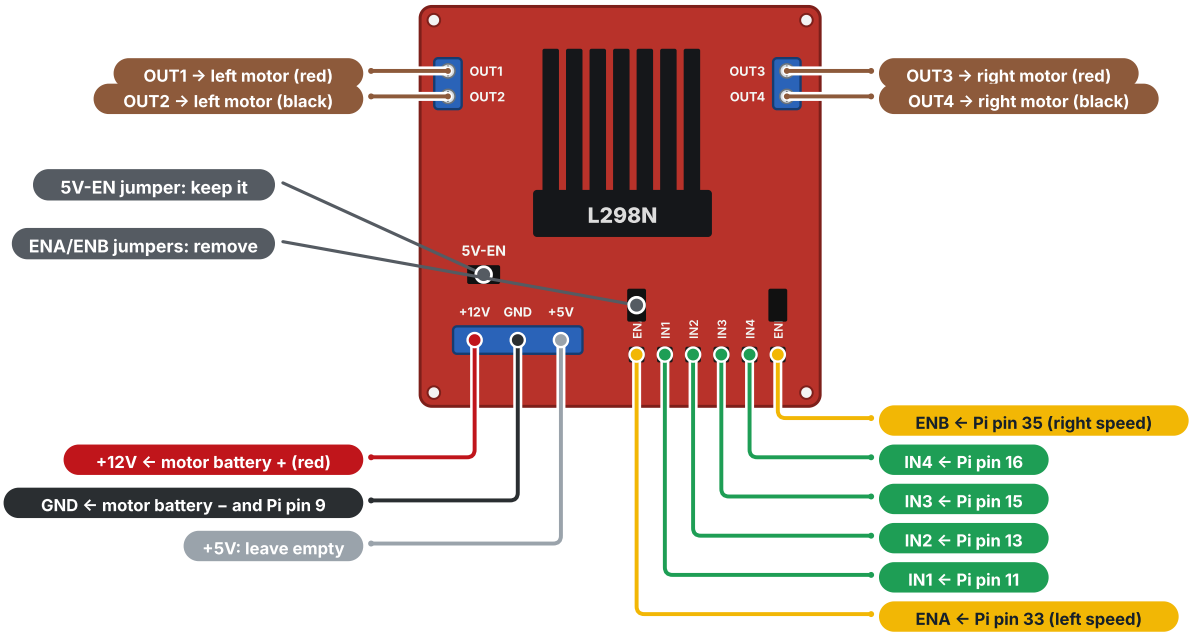
Pin numbers are counted on the board (1-40). Coloured pins are used by the robot; grey ones are free.



The 40-pin header, top view, USB ports pointing down the page.

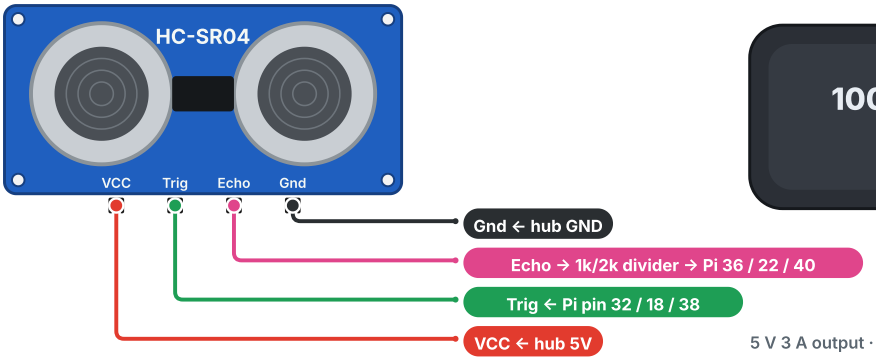
Board close-ups

L298N motor driver: every terminal and where it goes

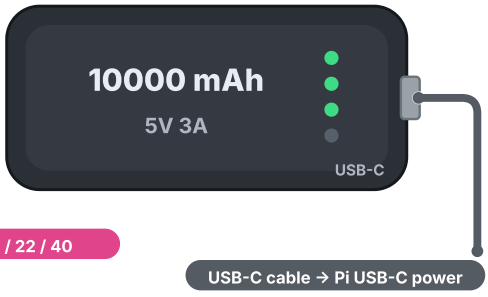


The L298N: battery and motors on the screw terminals, the Pi on the 6 pins.

HC-SR04 ultrasonic sensor

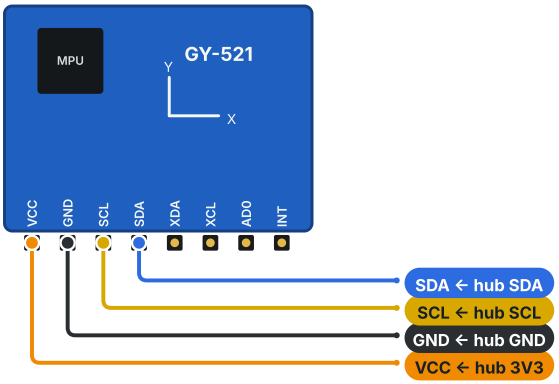


USB power bank → Raspberry Pi

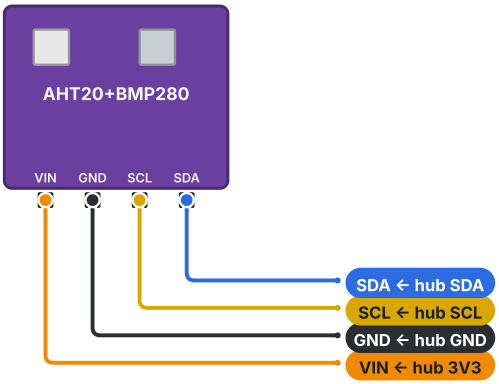


5 V 3 A output · the Pi only · LEDs show the charge

MPU6050 (GY-521)



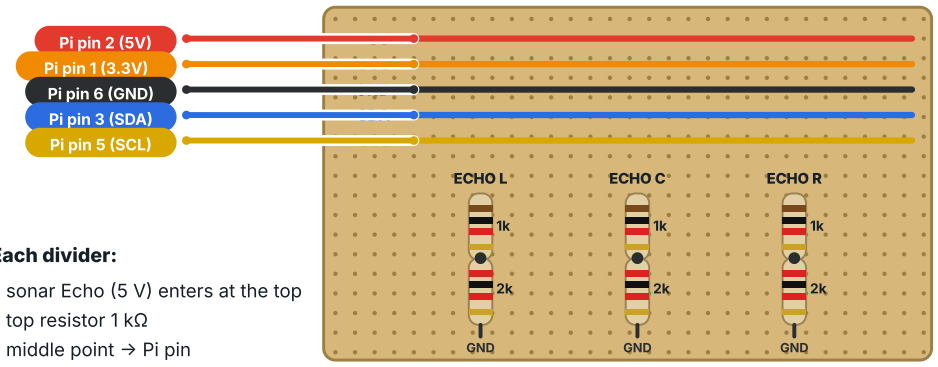
AHT20 + BMP280



MPU6050 = I²C address 0x68 · AHT20 = 0x38, BMP280 = 0x77 (or 0x76). The other GY-521 pins are not used.

HC-SR04, power bank, MPU6050 and AHT20: every pin and where it goes.

The hub board (you make it)



Each divider:

- sonar Echo (5 V) enters at the top
- top resistor 1 kΩ
- middle point → Pi pin
- bottom resistor 2 kΩ → GND
- 5 V × 2/3 = 3.3 V: safe

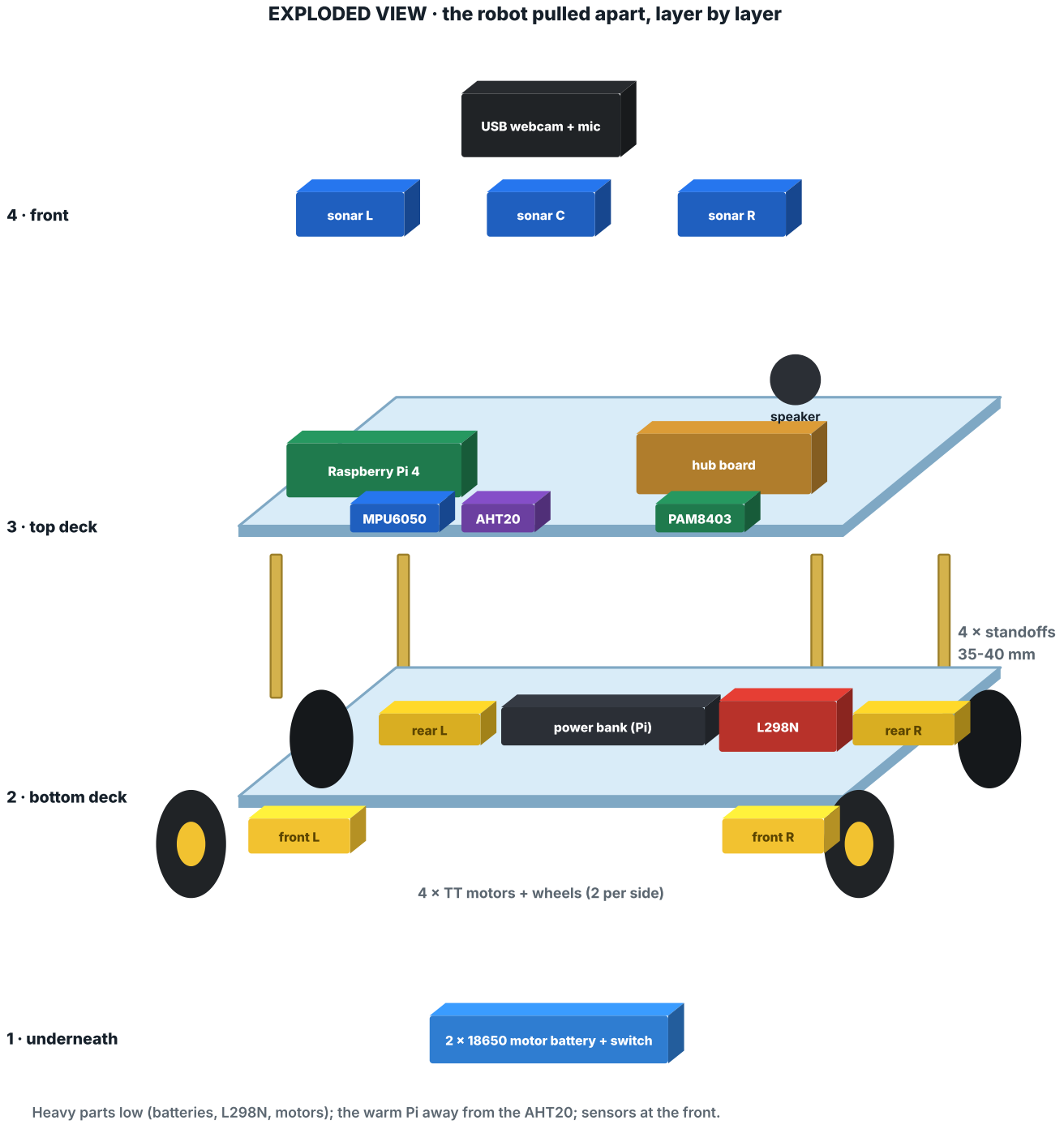
Solder the rails as bare wire along the strips;
every module plugs into the rails with jumper wires.

Five rails on top, the three sonar Echo dividers below.

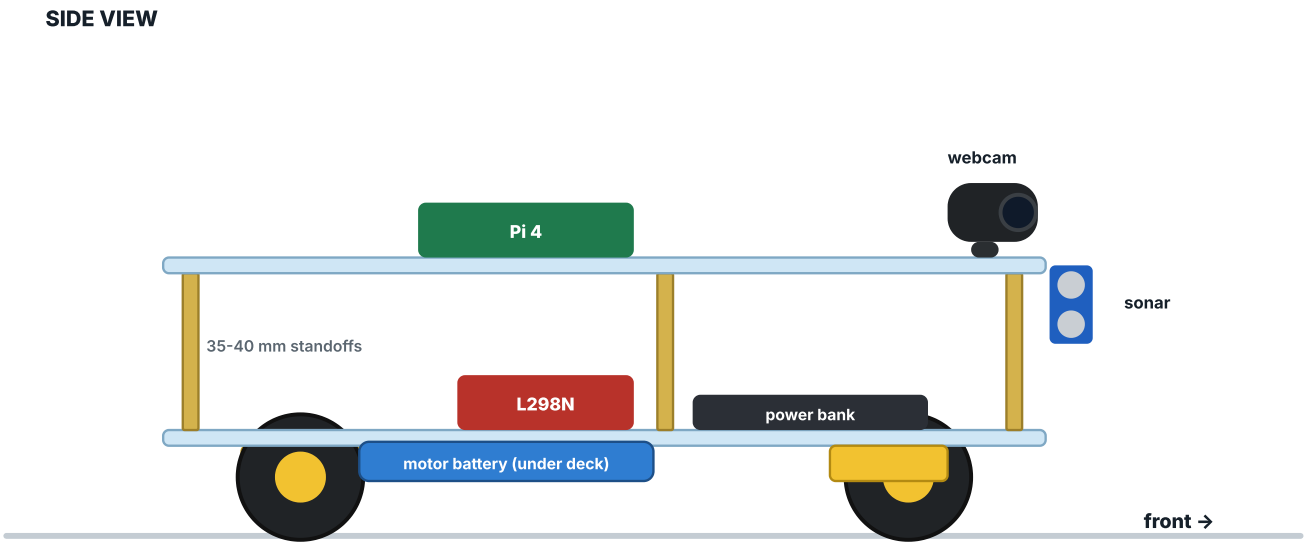
The hub board saves wires: the Pi gives 5 V, 3.3 V, GND, SDA and SCL to the hub once, and every sensor plugs into the hub. The dividers turn the sonars' 5 V Echo signals into a safe 3.3 V.

C The robot's structure

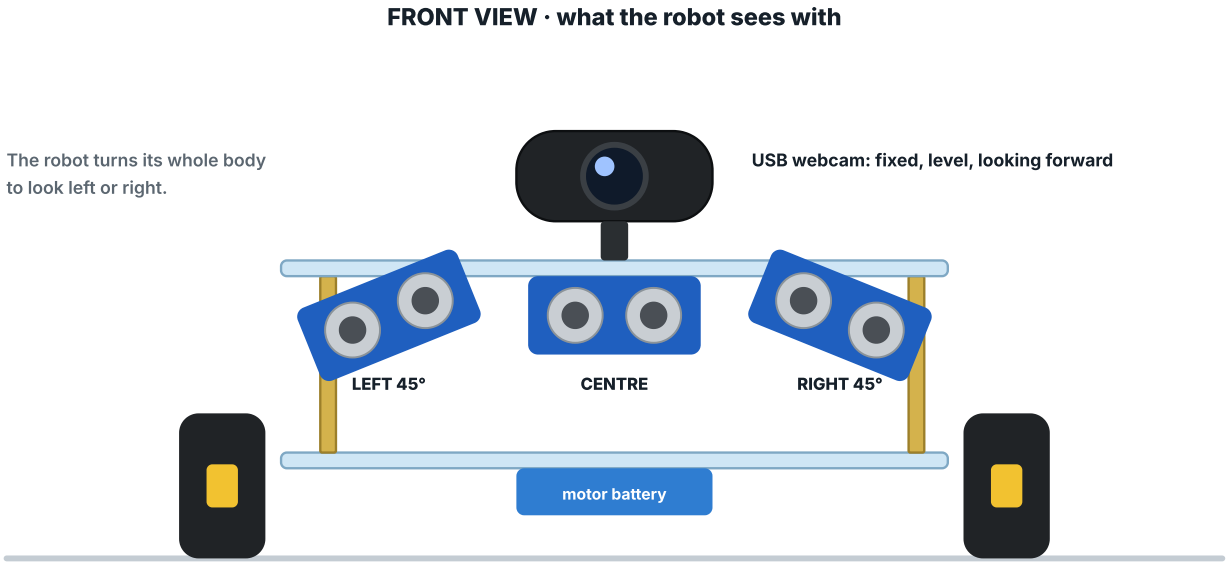
Two acrylic decks on standoffs. The motor battery hangs underneath, the motors, the power bank and the motor driver sit on the bottom deck, the computer and sensors on the top deck, and the distance sensors and webcam face forward.



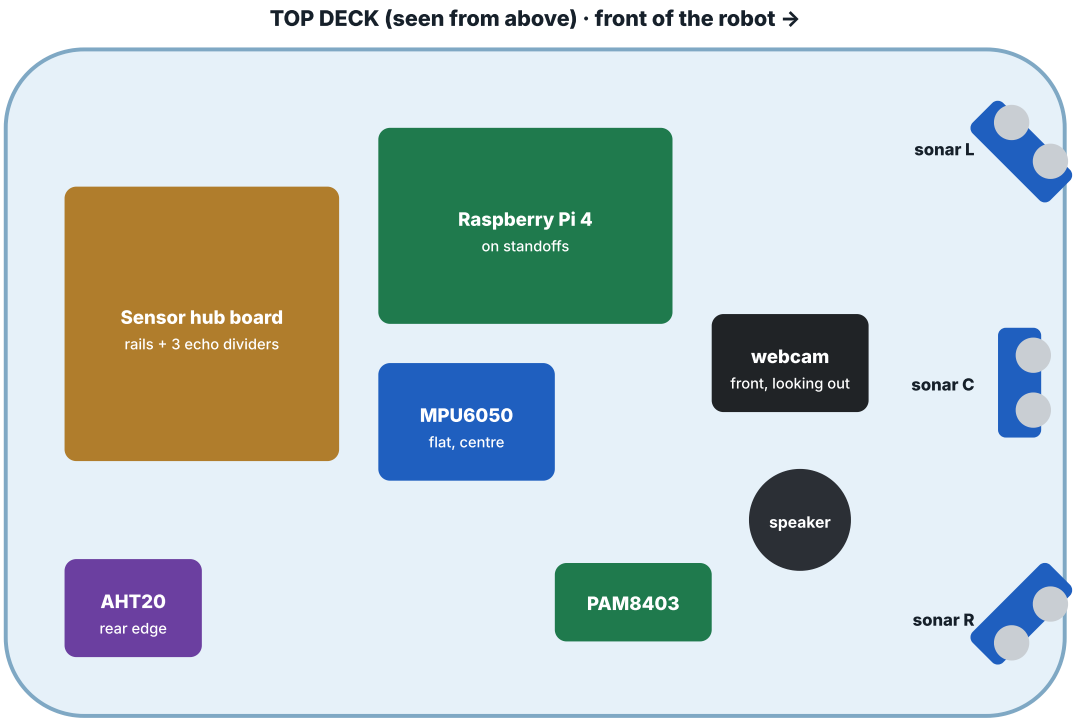
Exploded view: 1 motor battery, 2 bottom deck, 3 top deck, 4 front sensors.



Side view.

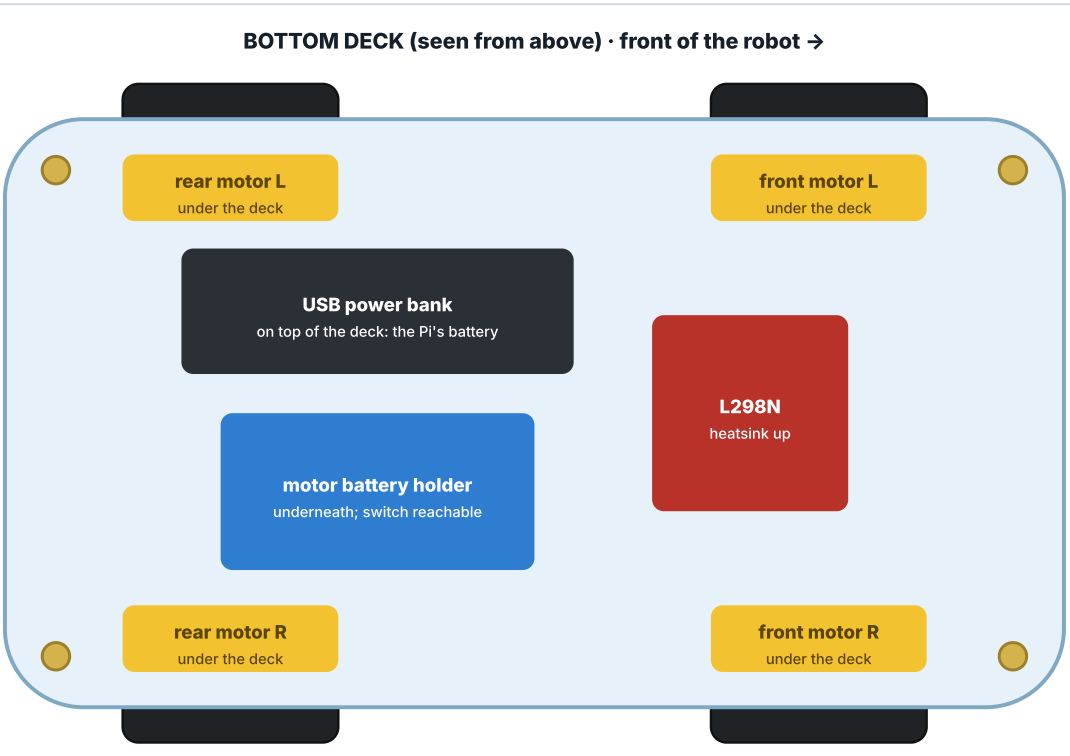


Front view: three distance sensors fanned out, the webcam above.



Bottom deck: power bank and L298N; the 4 motors and the motor battery underneath. Keep the AHT20 away from the Pi CPU.

Top deck.

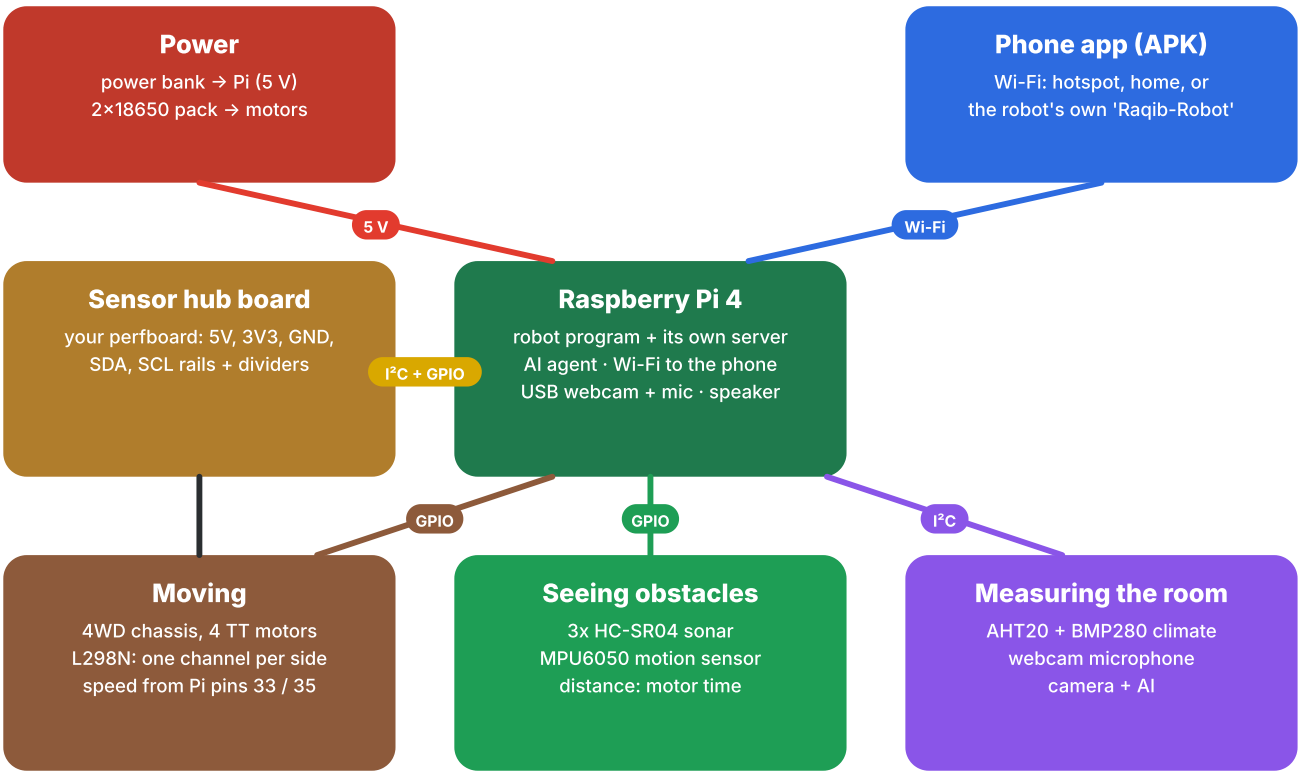


Gold dots = standoffs to the top deck. The USB-C cable, motor and battery wires go up through the deck holes.

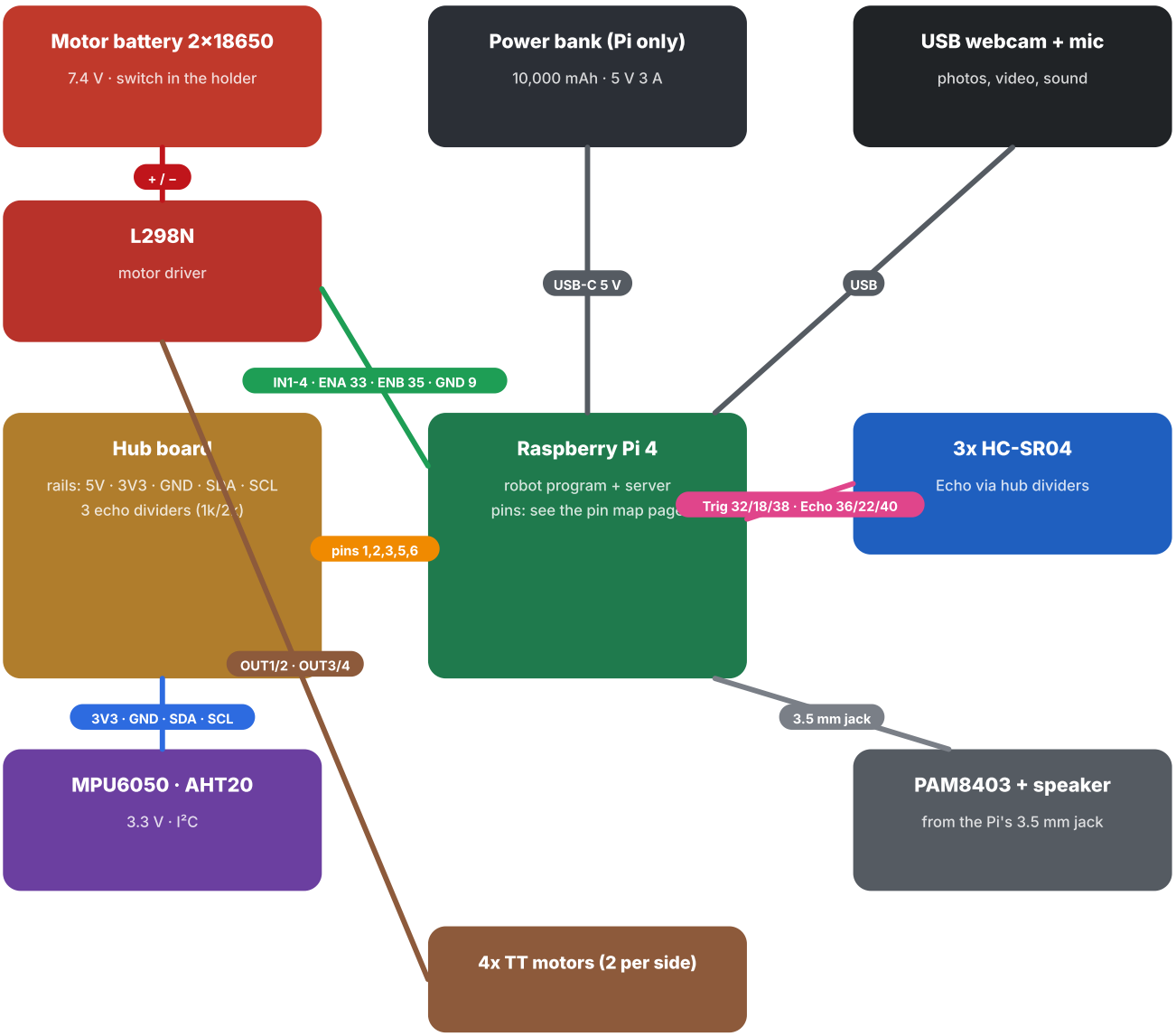
Bottom deck.

D How everything connects

Start with the big picture, then one diagram per group. The pin numbers are the physical pins of the Raspberry Pi header.



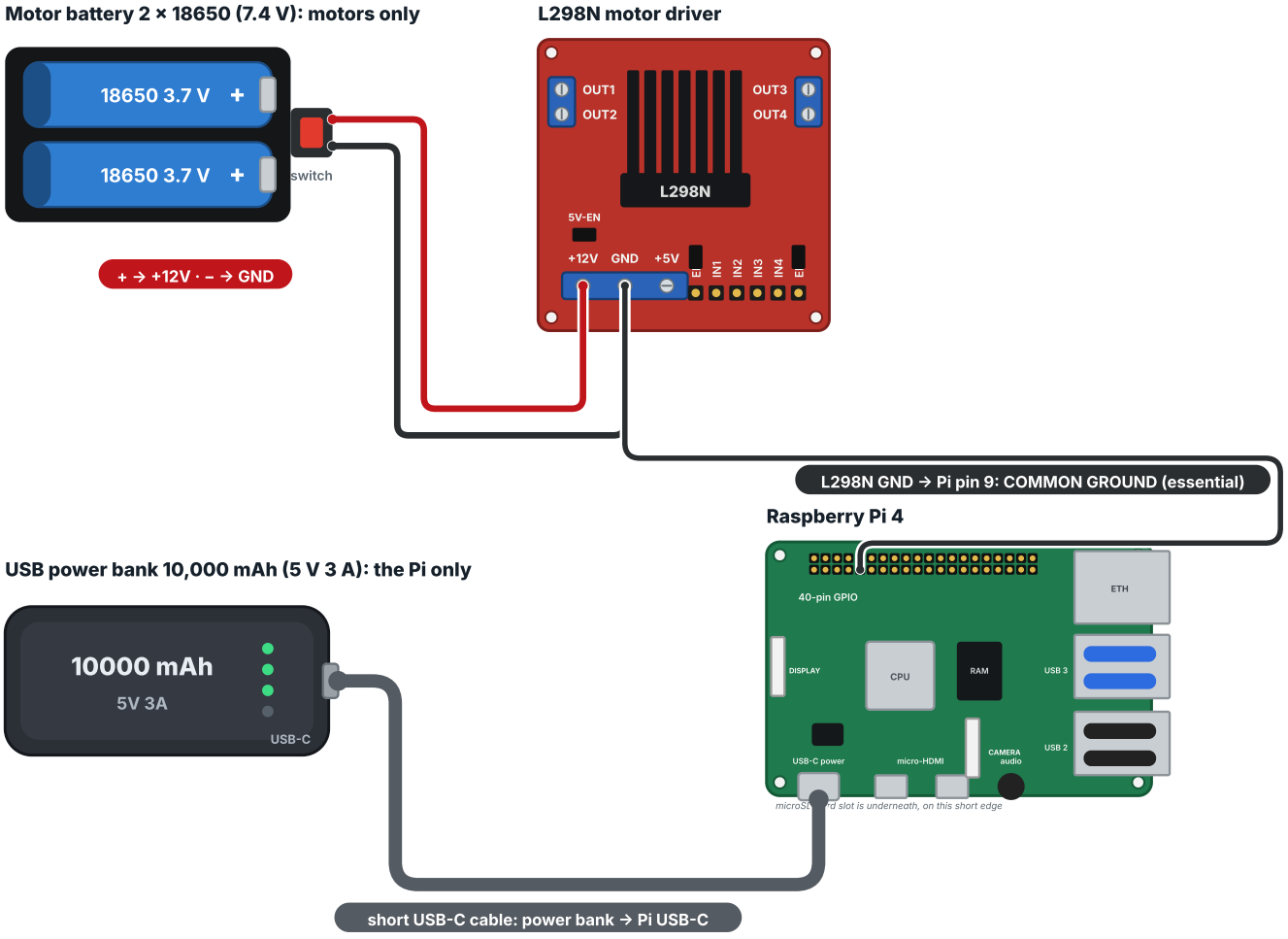
The system: five groups of parts around the Raspberry Pi.



Two separate batteries; L298N GND → Pi pin 9 joins their grounds. Sonar Echo → hub divider → Pi.

The complete harness: every block and the wires between them.

1 · Power

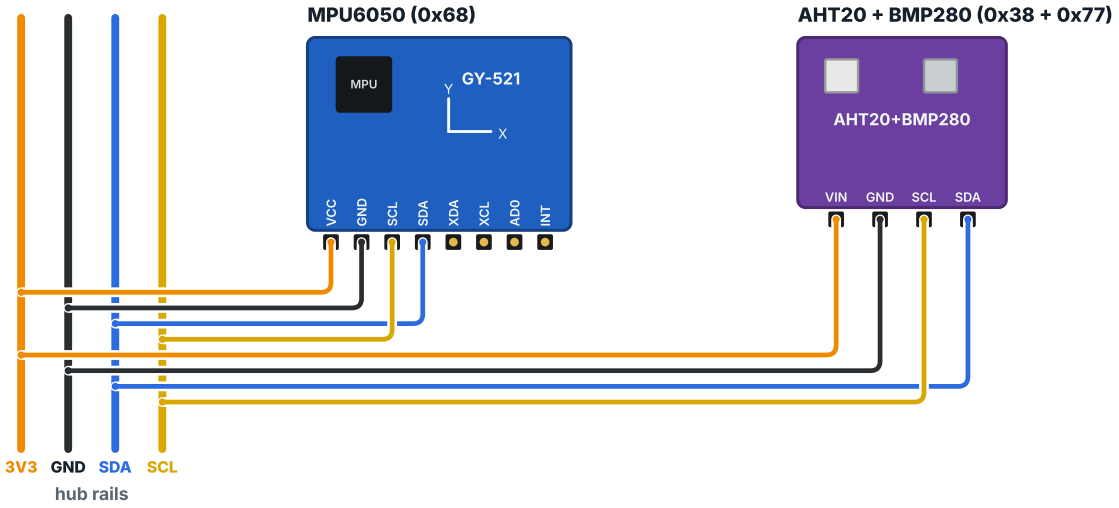


Two batteries: motor current spikes can never restart the Pi, and there is no voltage to set by hand.

The one wire that joins the two systems is the common ground; without it the motors ignore the Pi.

Power bank → Pi USB-C; 2 × 18650 → L298N; L298N GND → Pi pin 9 (common ground).

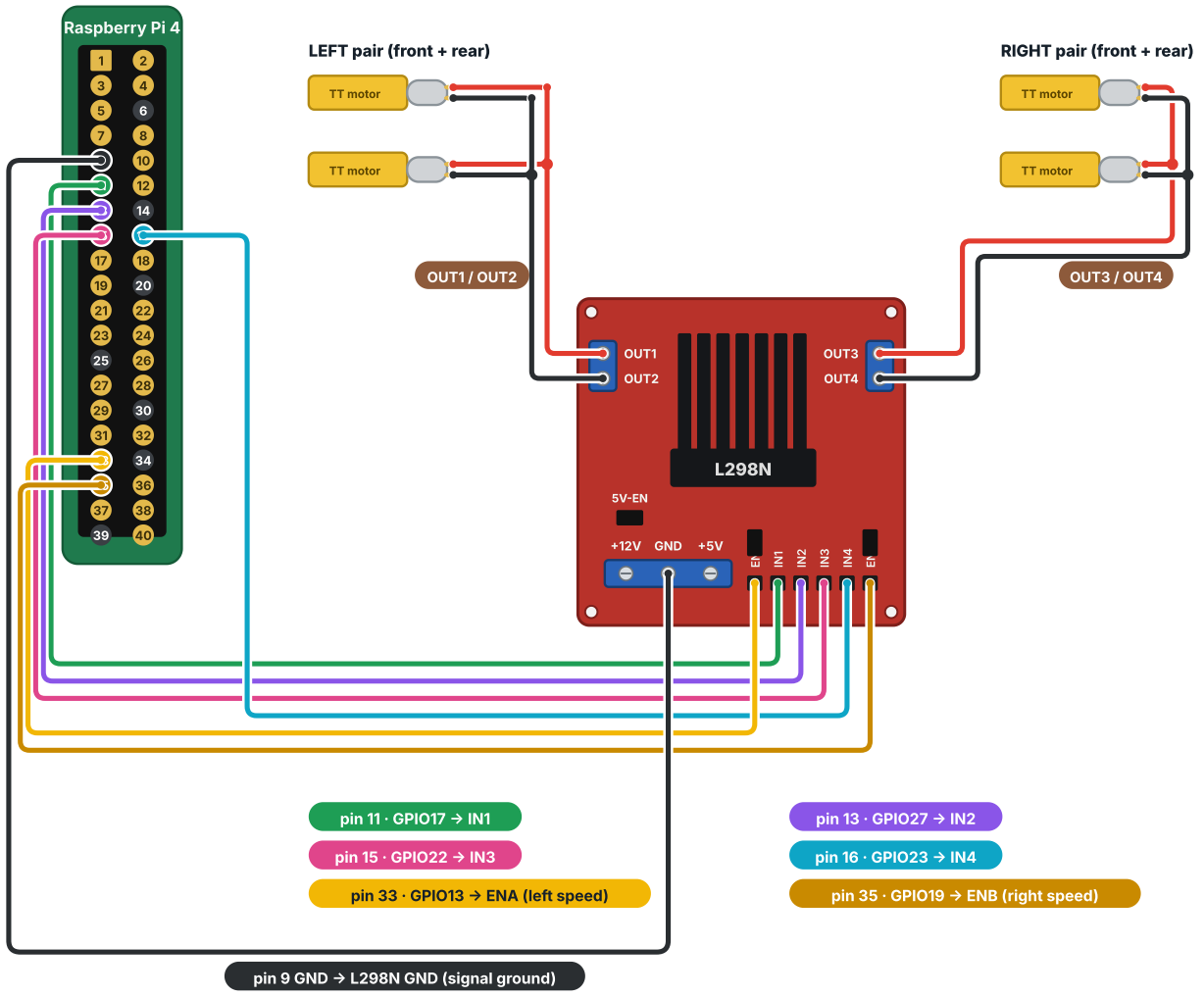
2 · I²C sensors



- Both boards join the same four rails: 3.3 V, GND, SDA, SCL (Pi pins 1, 6, 3, 5).
- Each answers to its own address: MPU6050 0x68, AHT20 0x38, BMP280 0x77 (or 0x76).
- Test: python -m hwtest.i2c_scan lists every address found.

MPU6050 and AHT20 + BMP280 share four wires.

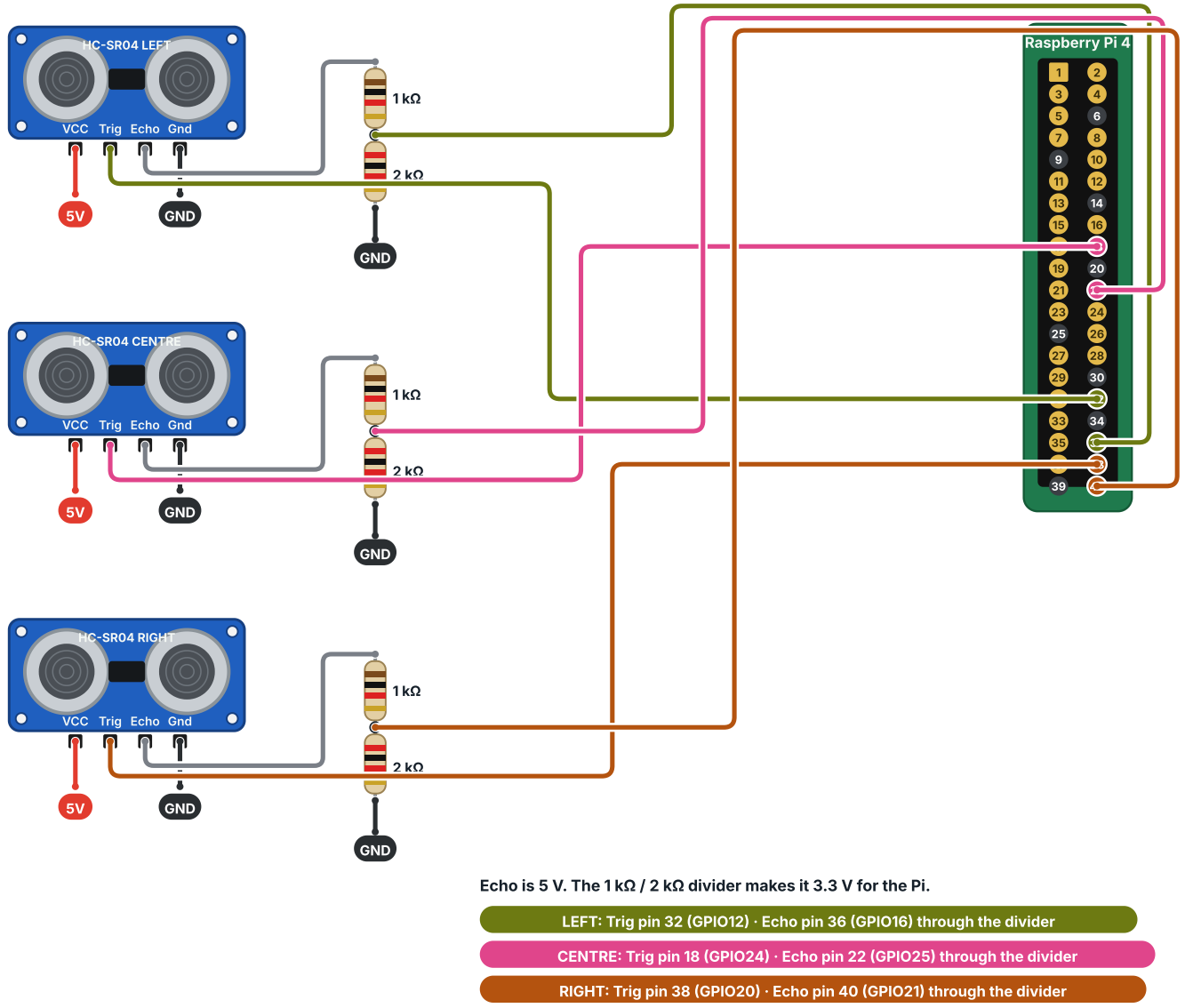
3 · Motors



- Pull the ENA and ENB jumpers off: the Pi sets the speed on pins 33 and 35. Keep the 5V-EN jumper.
- +12V and GND come from the battery (power page). Leave the L298N's +5V terminal unconnected.
- Each side's two motors are joined (red to red, black to black) and run together: skid steering.

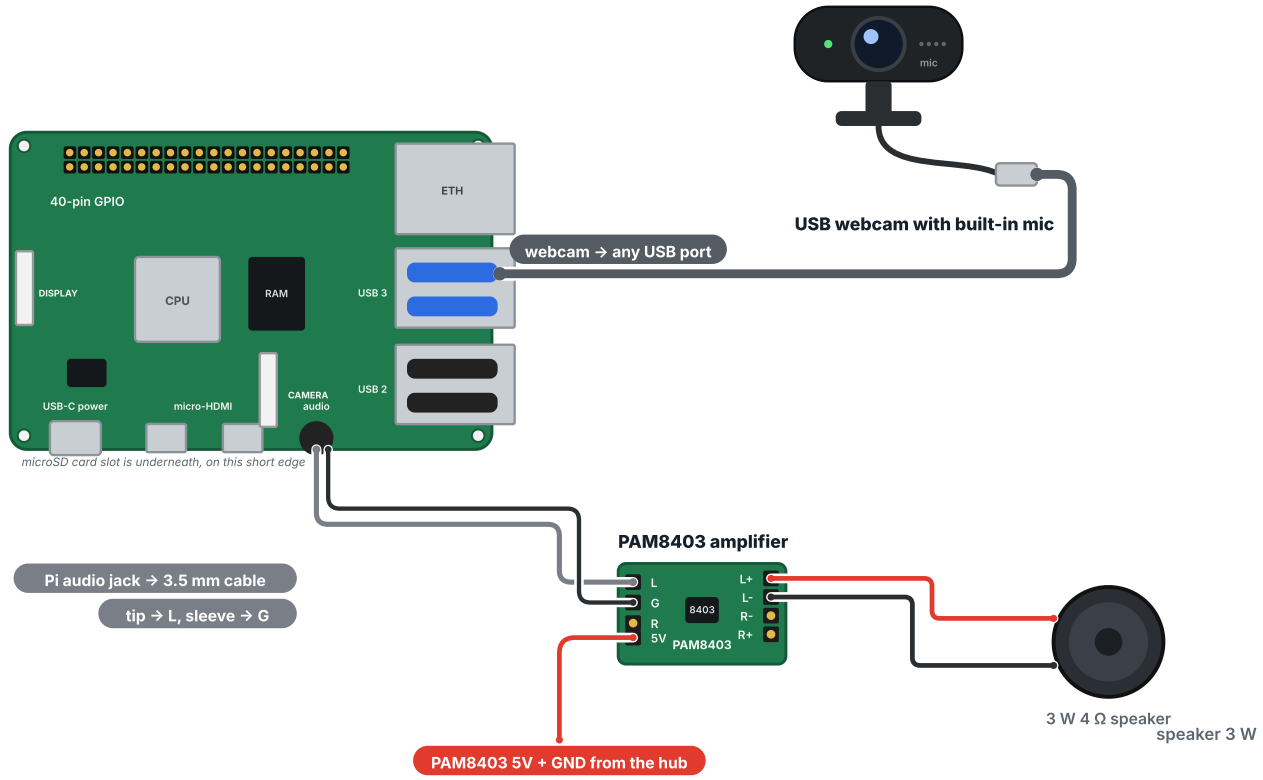
Direction on pins 11/13/15/16, speed on pins 33/35.

4 · Distance sensors



Each Echo goes through a 1 kΩ / 2 kΩ divider.

5 · Camera, microphone and speaker



Webcam on USB; Pi audio jack → PAM8403 → speaker.

Every wire, one by one

The same list the robot software and the app's Wiring Bench use. Tick each wire off as you connect it.

Part	Pin	Goes to	Wire
USB power bank (Pi only)	**USB-C cable**	USB-C power input	USB cable
L298N motor driver	**+12V**	Battery +	red
	GND (battery)	Battery -	black
	GND (to Pi)	GND (hub GND rail / any Pi GND pin)	black
	ENA	Pi pin 33 (GPIO13)	any colour
	IN1	Pi pin 11 (GPIO17)	any colour
	IN2	Pi pin 13 (GPIO27)	any colour
	ENB	Pi pin 35 (GPIO19)	any colour
	IN3	Pi pin 15 (GPIO22)	any colour
	IN4	Pi pin 16 (GPIO23)	any colour
TT gear motors (4)	**Left pair**	OUT1 + OUT2	motor wires
	Right pair	OUT3 + OUT4	motor wires
HC-SR04 ultrasonic - Left	**VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	TRIG	Pi pin 32 (GPIO12)	any colour
	ECHO	Pi pin 36 (GPIO16) · through its divider	any colour
HC-SR04 ultrasonic - Center	**VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	TRIG	Pi pin 18 (GPIO24)	any colour
	ECHO	Pi pin 22 (GPIO25) · through its divider	any colour
HC-SR04 ultrasonic - Right	**VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	GND	GND (hub GND rail / any Pi GND pin)	black
	TRIG	Pi pin 38 (GPIO20)	any colour
	ECHO	Pi pin 40 (GPIO21) · through its divider	any colour
MPU6050 IMU	**VCC**	3.3 V (hub 3V3 rail / Pi pin 1 or 17)	orange
	GND	GND (hub GND rail / any Pi GND pin)	black
	SDA	Pi pin 3 (GPIO2 SDA) · via hub SDA rail	blue
	SCL	Pi pin 5 (GPIO3 SCL) · via hub SCL rail	yellow
AHT20 + BMP280 climate sensor	**VIN**	3.3 V (hub 3V3 rail / Pi pin 1 or 17)	orange
	GND	GND (hub GND rail / any Pi GND pin)	black
	SDA	Pi pin 3 (GPIO2 SDA) · via hub SDA rail	blue
	SCL	Pi pin 5 (GPIO3 SCL) · via hub SCL rail	yellow
USB webcam with microphone	**USB plug**	USB port	USB cable
Speaker + PAM8403 amp	**Amp VCC**	5 V (hub 5V rail / Pi pin 2 or 4)	red
	Amp GND	GND (hub GND rail / any Pi GND pin)	black
	Audio in (L)	3.5 mm audio jack	audio cable